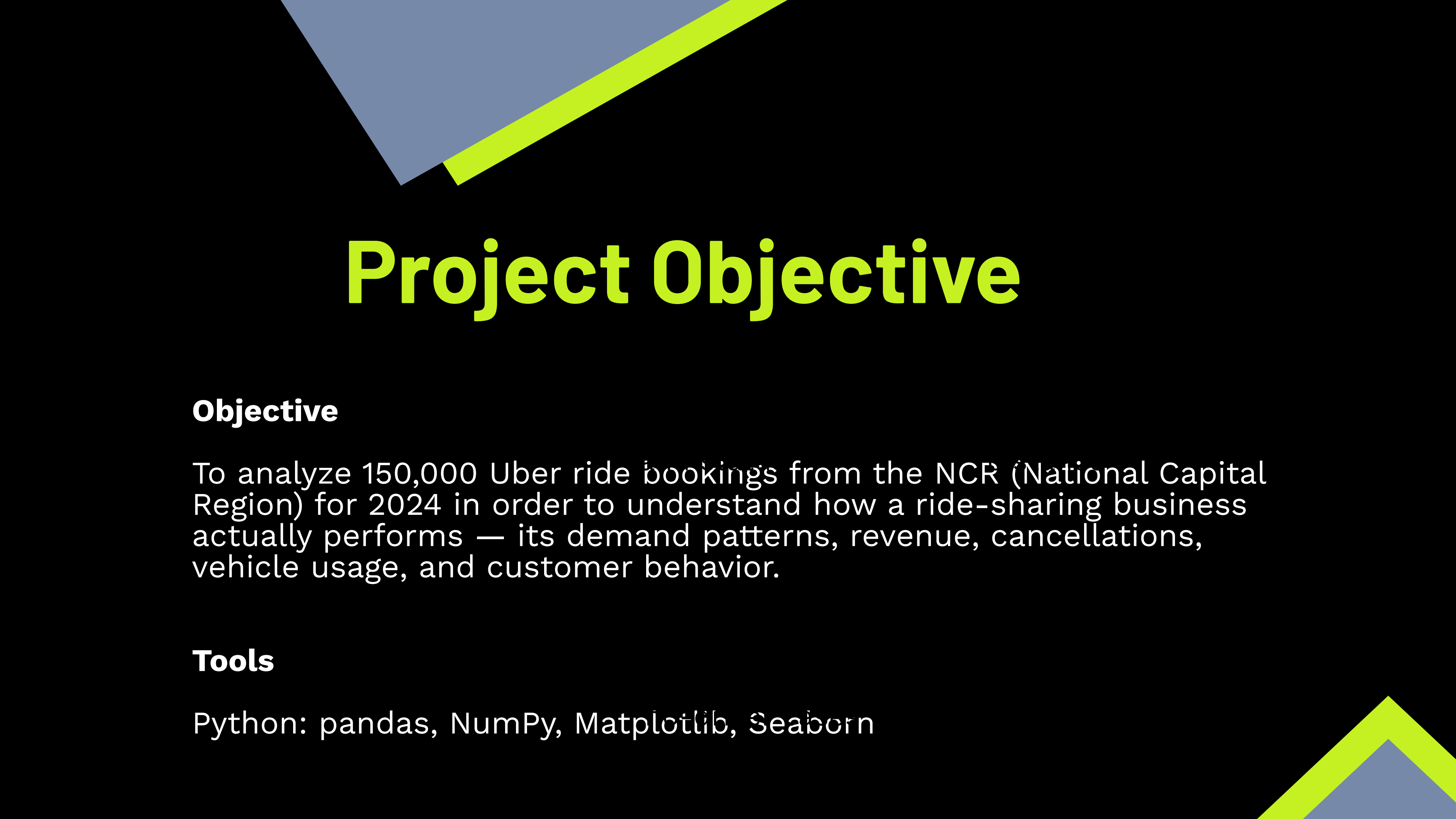


RidePulse

Ride-Sharing Revenue & Operations Intelligence

NCR Ride Bookings • Executive Analytics Review





Project Objective

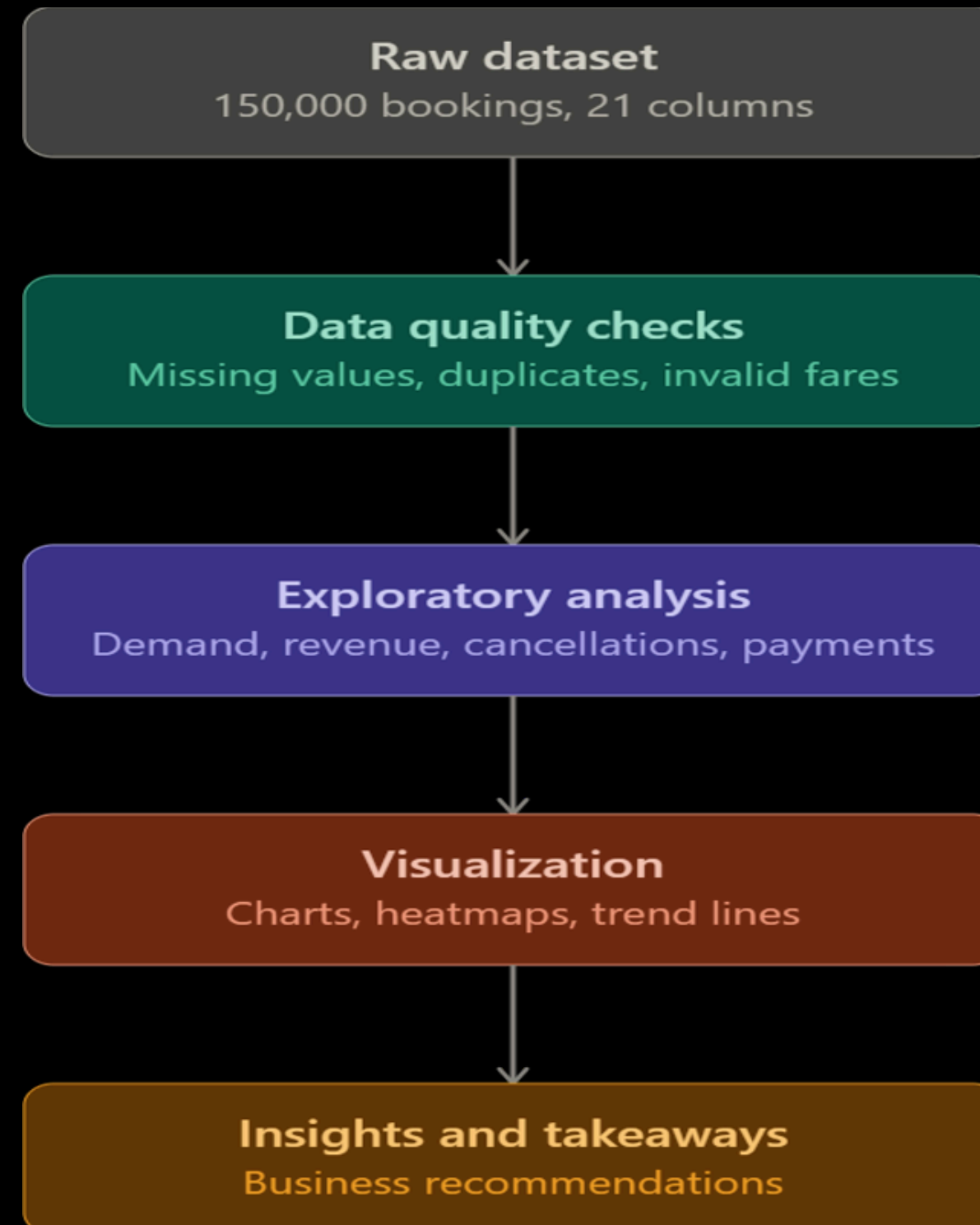
Objective

To analyze 150,000 Uber ride bookings from the NCR (National Capital Region) for 2024 in order to understand how a ride-sharing business actually performs — its demand patterns, revenue, cancellations, vehicle usage, and customer behavior.

Tools

Python: pandas, NumPy, Matplotlib, Seaborn

WorkFlow



Understanding the dataset

150,000

Booking records

21

Columns

176

Unique pickup locations

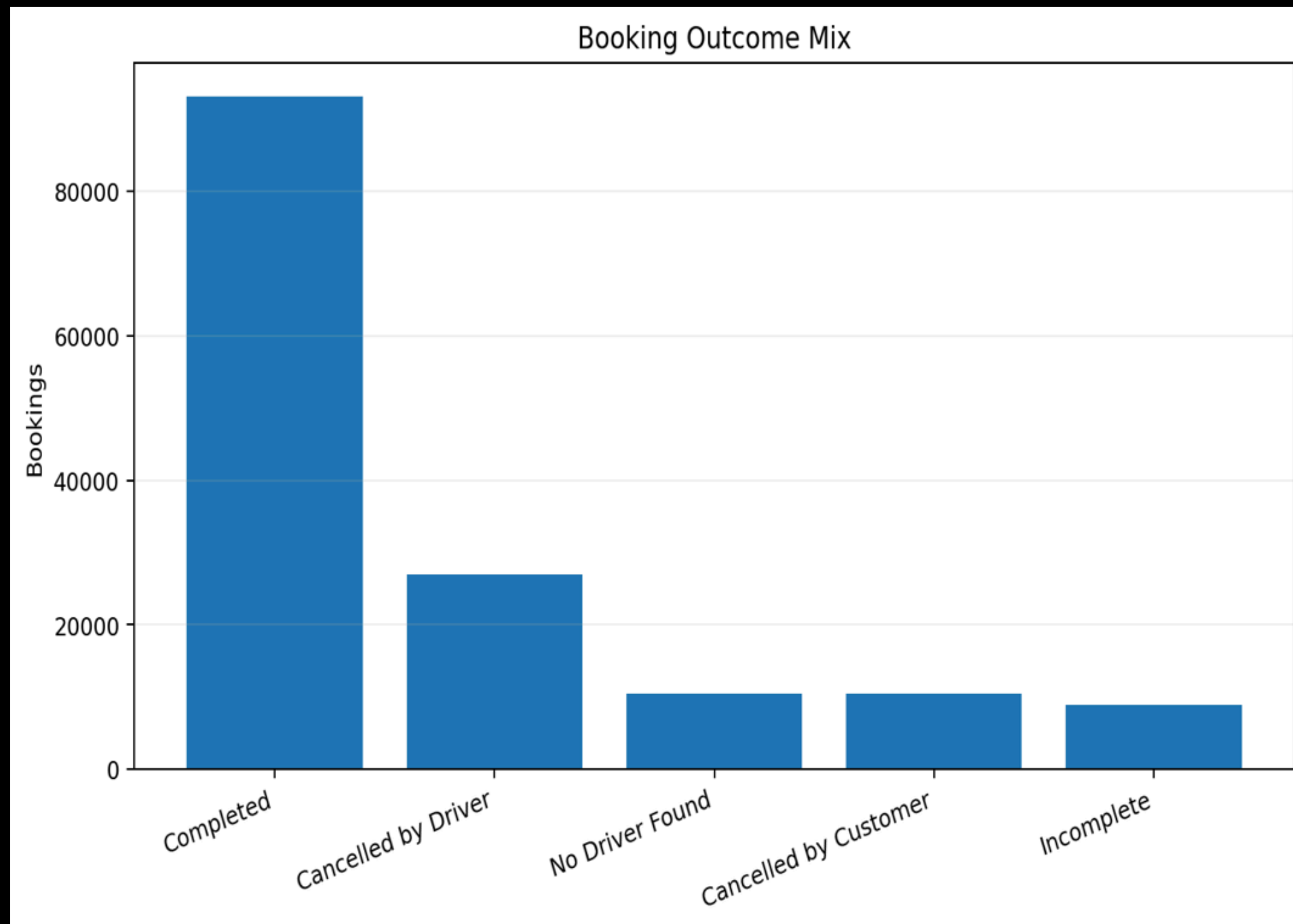
2024

Jan 1 to Dec 30

Column groups

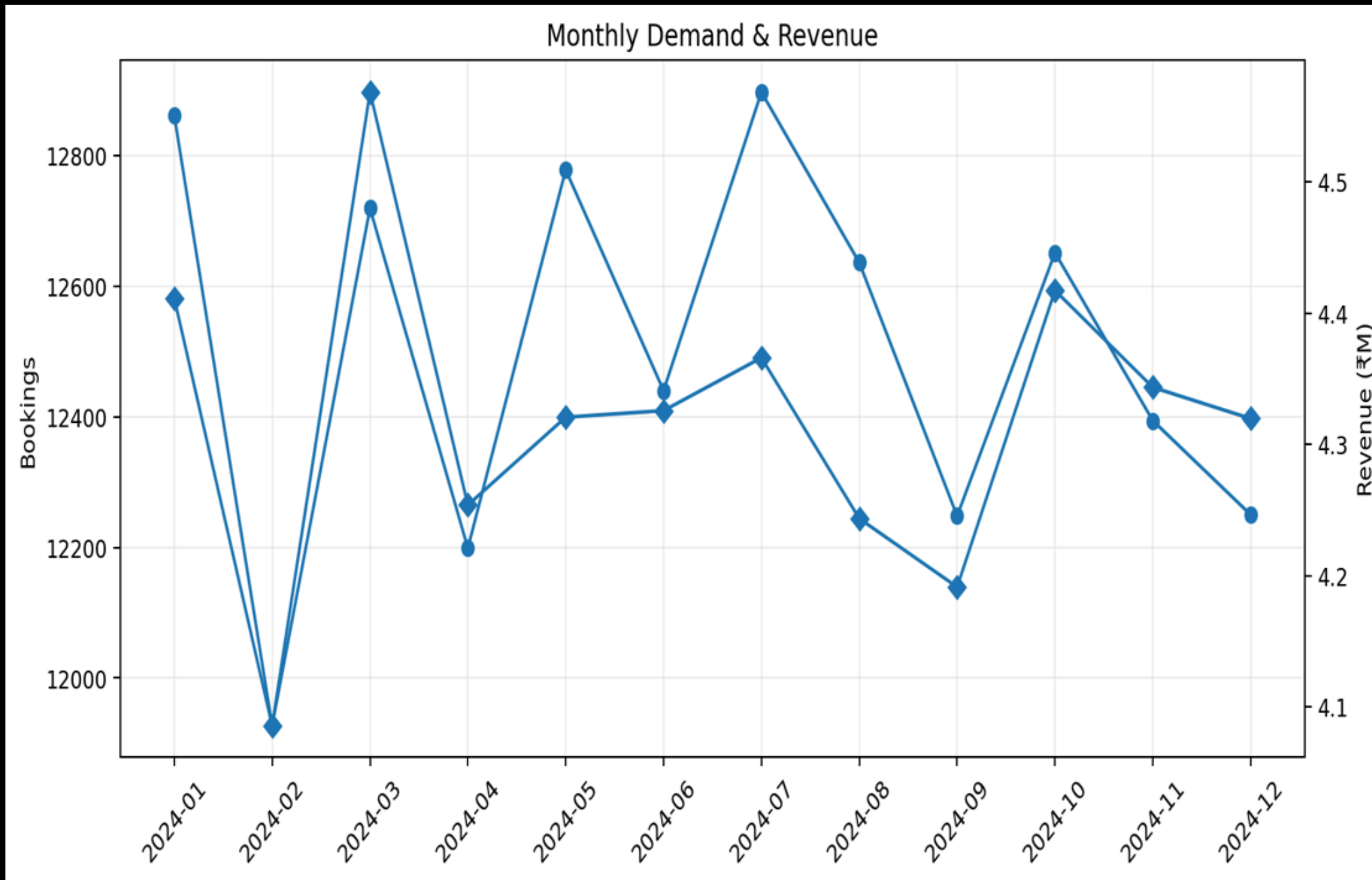
- Booking: date, time, booking ID, status, customer ID
- Trip: vehicle type, pickup and drop location, VTAT and CTAT
- Outcome: cancellation and incomplete-ride reasons, booking value, distance, ratings, payment method

Data Quality Checks



- Most rides go fine — about 62% are completed.
- The biggest problem is drivers cancelling — that's 18% of all bookings.
- A smaller chunk fail because no driver was found, or the customer cancelled — about 7% each.
- A few rides start but don't finish — around 6%.
- So roughly 1 in 3 rides has some kind of problem, and most of that comes from the driver side, not the customer.

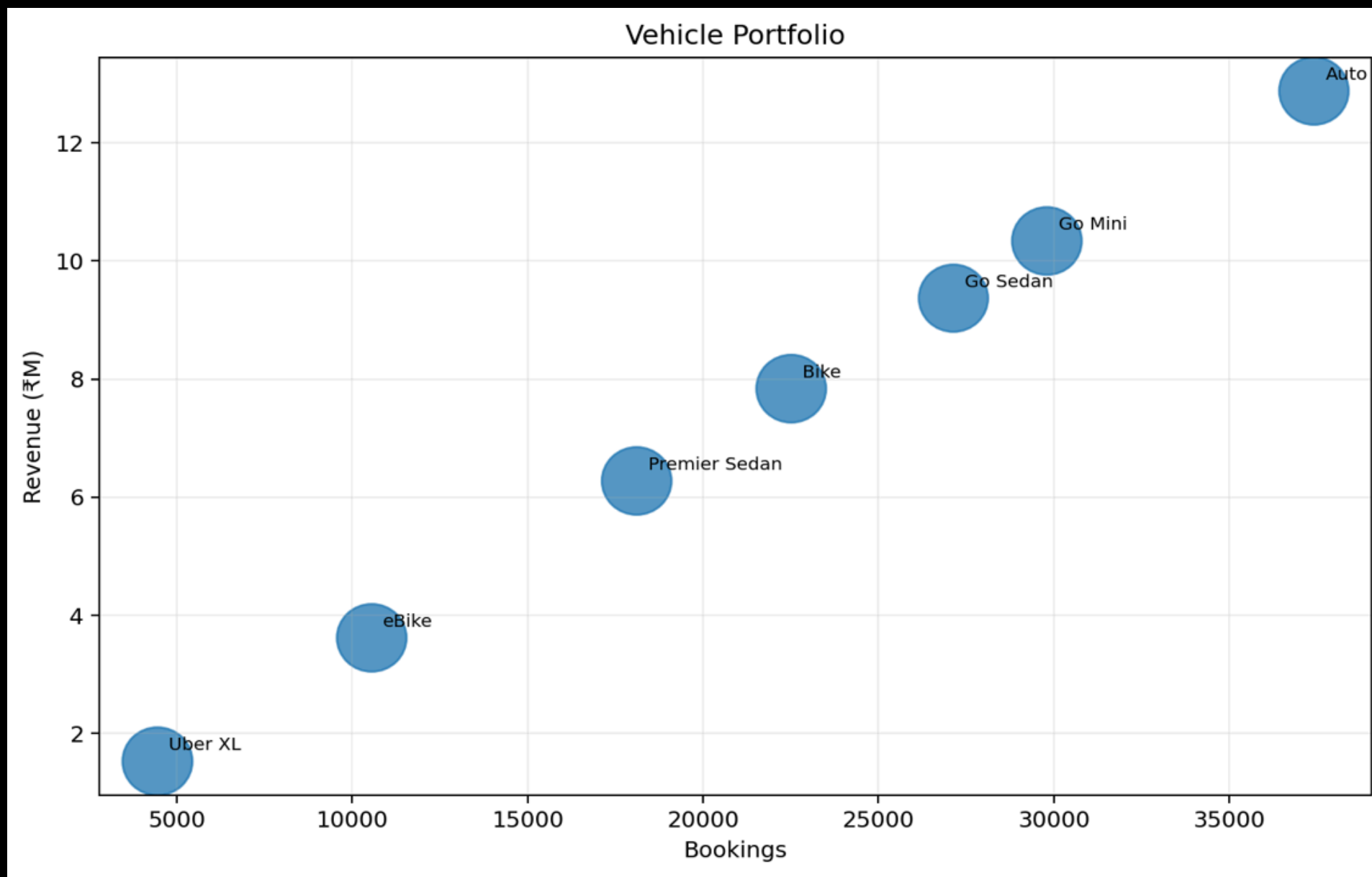
Demand & Revenue



This graph shows monthly bookings (demand) and revenue from January to December 2024.

Overall: Demand and revenue fluctuate throughout the year, with July showing the highest demand and February showing the lowest demand.

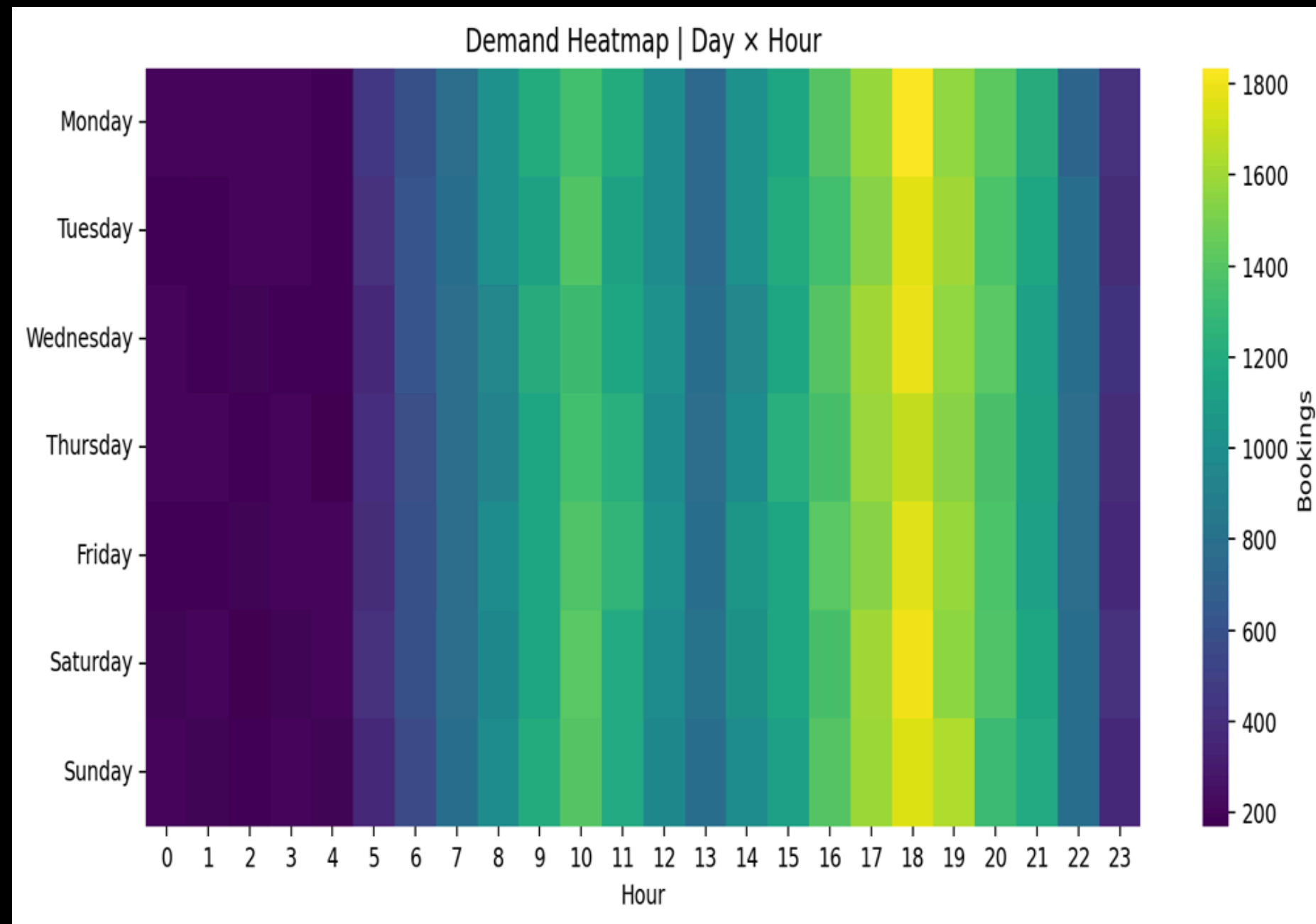
Bookings and Revenue by Vehicle Type



This graph shows the relationship between bookings and revenue for different vehicle types.

Overall: As the number of bookings increases, revenue generally increases. Auto has the strongest overall performance in this

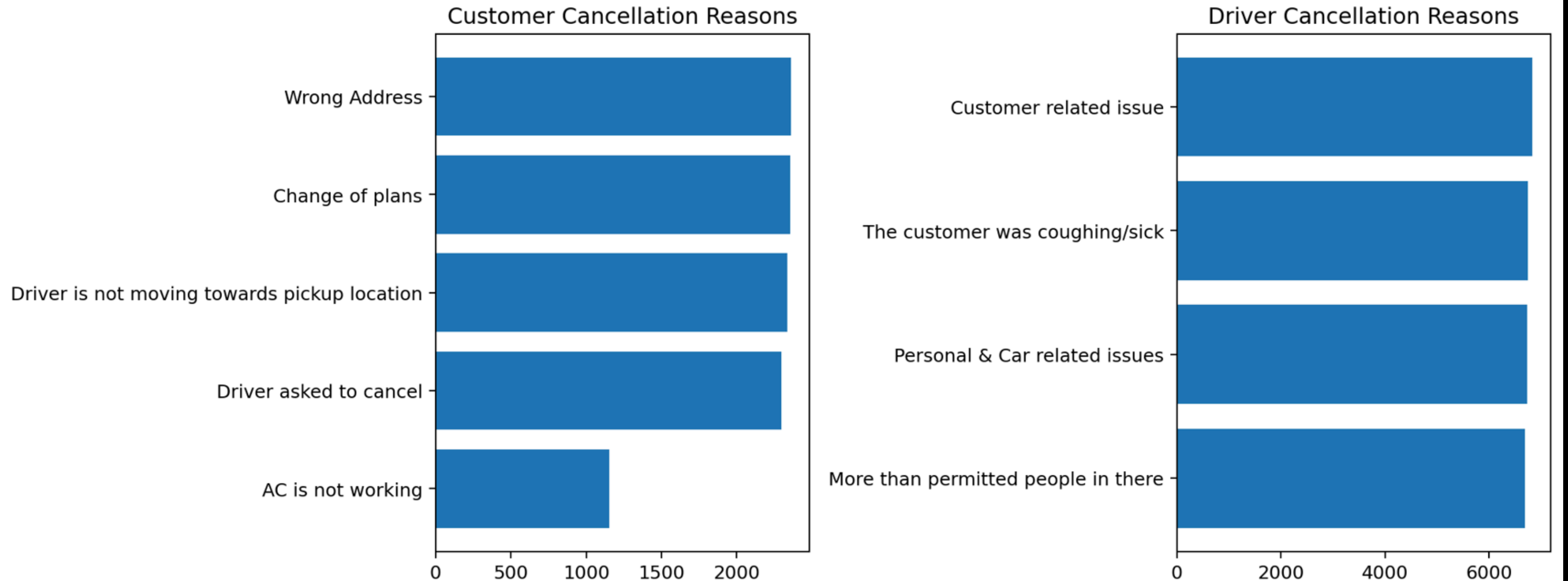
Data Visualization



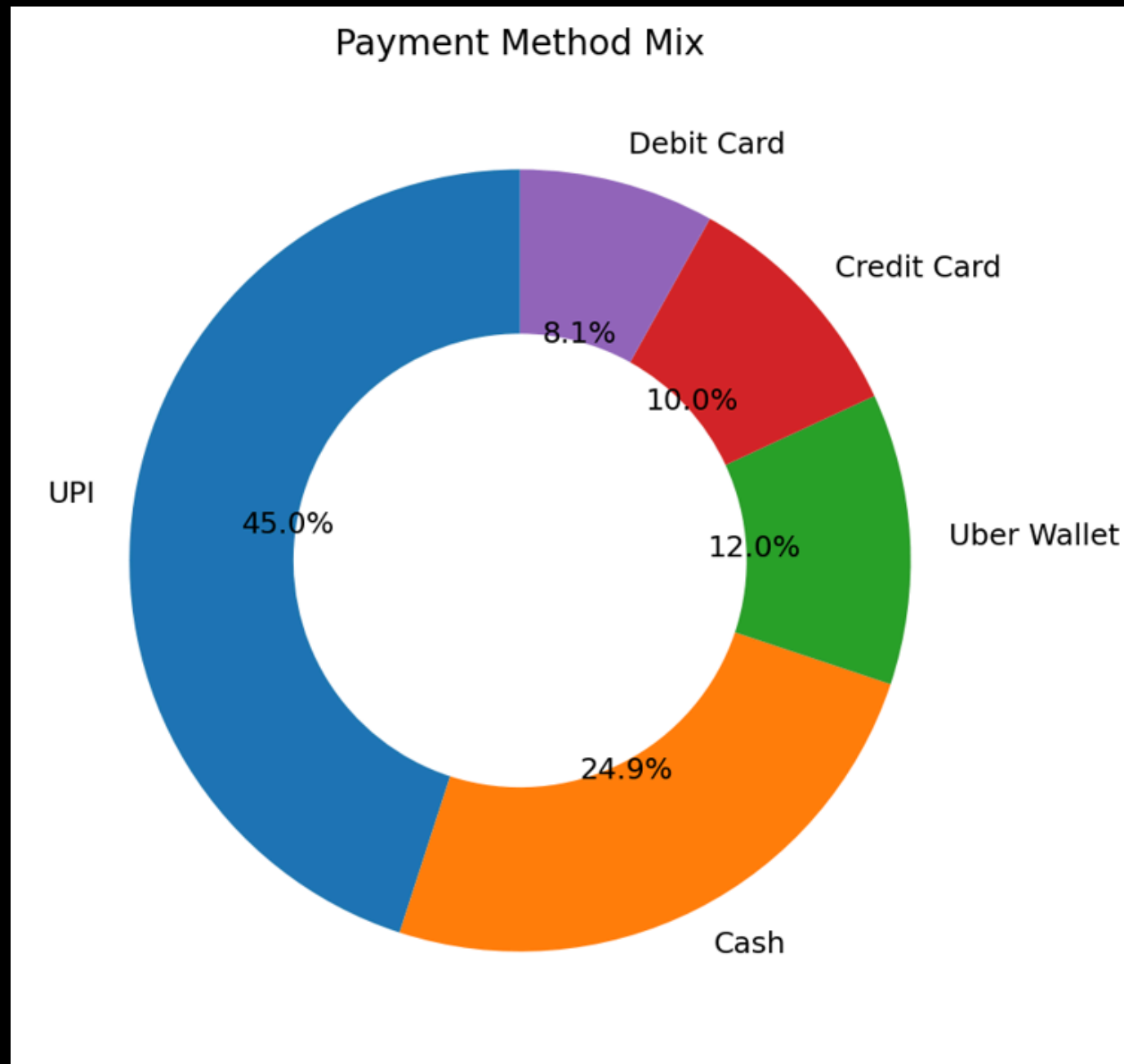
- The busiest time is around 6–7 PM (18:00–19:00) — clearly the peak, on every day of the week.
- Rides are lowest late night and early morning, roughly midnight to 5 AM.
- Demand builds steadily through the morning, stays fairly strong all afternoon, then peaks in the evening.
- This pattern is nearly identical across all 7 days — weekdays and weekends look the same, no strong "weekend effect."
- Bottom line: demand is driven mainly by time of day, not day of the week — evening rush hour is the key period to focus on.

Root Causes

Operational Leakage Root Causes



Payment Behavior

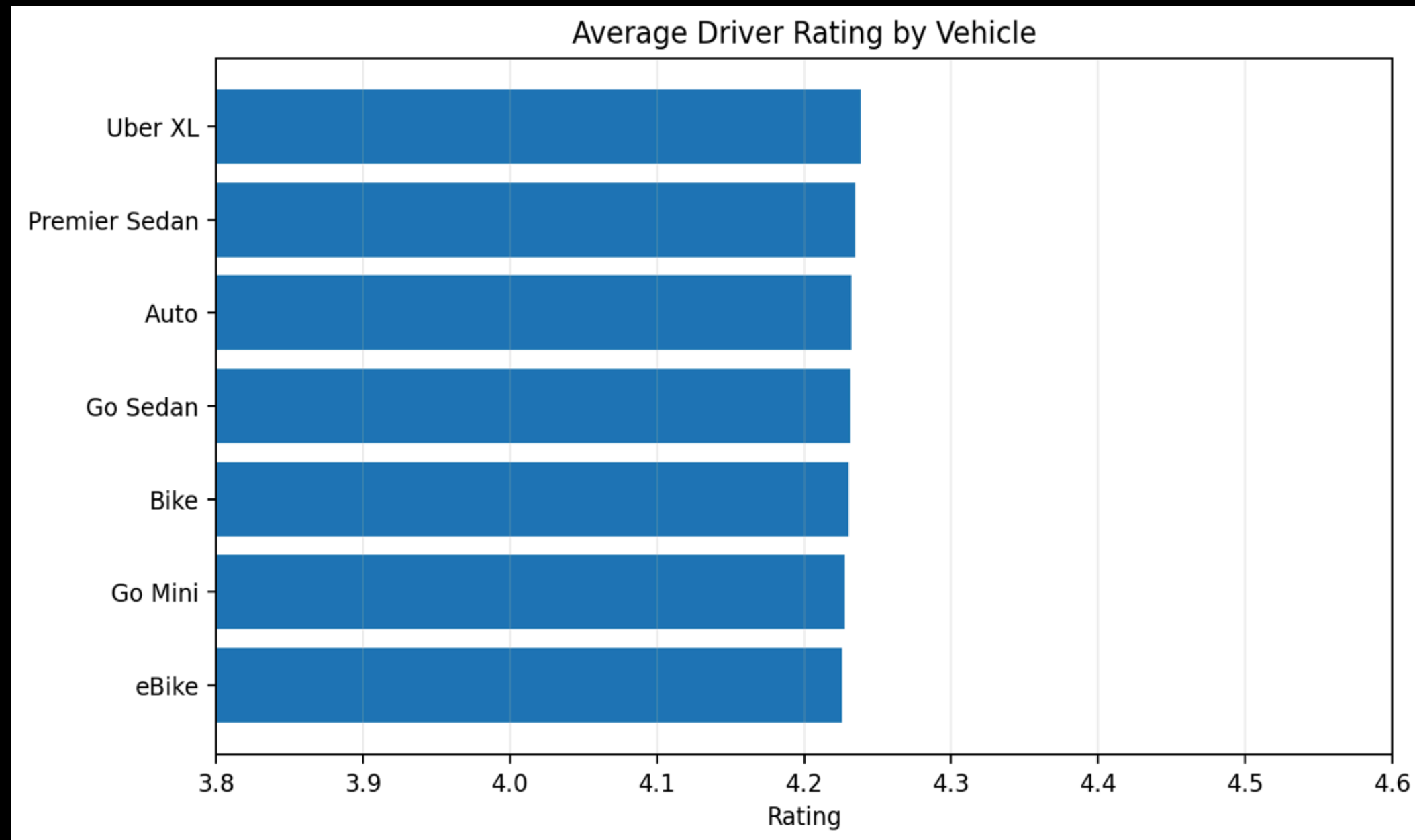


UPI is the most-used recorded payment method at 45.0% of records with a payment method.

Cash represents 24.9% of all records.

Digital payment behavior can be monitored alongside completion and revenue to identify conversion patterns.

Driver Ratings By Vehicle Type



Overall average driver rating: 4.23.

Overall average customer rating: 4.40.

Fare, Distance & Ratings

Rs 508

Average booking value

Rs 414

Median booking value

24.6 km

Average ride distance

4.23

Average driver rating

Reading the numbers

- Fares range from Rs 50 to Rs 4,277; mean above median means a few long or premium trips pull the average up.
- Distances span 1 to 50 km, spread evenly (median 23.7 km).
- Driver ratings sit between 3.0 and 5.0, so service quality is consistently good.

Key Takeaways

62%

of bookings complete; 38% are lost to cancellation, no driver or incomplete trips

18%

driver cancellations, the top lever for improving completion

63%

of demand from Auto, Go Mini and Go Sedan

45%

of payments via UPI, with cash at 25%

Conclusion

- RidePulse has healthy demand, but the main issue is completing rides due to driver shortages and cancellations. The evening peak, especially 6–7 PM, is the most critical period. Improving driver availability and reducing cancellations can increase revenue more than demand generation.